MA4142: Regression and Multivariate Analysis
Life Expectancy

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Table of Contents

1. About the Data
2. Exploratory Data Analysis
3. Simple Linear Regression
4. Multiple Linear Regression
5. Diagnostics and Remedy

About the Data

Dataset Introduction

- The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries
- The data-sets are made available to public for the purpose of health data analysis.
- The data-set related to life expectancy, health factors for 193 countries has been collected from the same WHO data repository website and its corresponding economic data was collected from United Nation website.
- Among all categories of health-related factors only those critical factors were chosen which are more representative.

More about the Data

- It has been observed that in the past 15 years , there has been a huge development in health sector resulting in improvement of human mortality rates especially in the developing nations in comparison to the past 30 years.
- The data is from year 2000-2015 for 193 countries for further analysis.
- The data was collected from WHO and United Nations website with the help of Deeksha Russell and Duan Wang.
- The data consists of 22 Columns and 2938 rows which meant 20 predicting variables.

Exploratory Data Analysis

Exploratory Data Analysis

Types of Features

For each country, in addition to a Status of Developed/Developing, we have:

Features related to Indicators of Country's Development :

Adult Mortality
Infant Deaths
GDP (per capita, in USD)
Population
Percentage Expenditure
Total Expenditure
Under-Five Deaths
Income Composition of Resources
Schooling

Features related to Individual Health :

Hepatitis.B
Diphtheria
HIV.AIDS
Measles
Polio
Alcohol
BMI
Thinness: 1-19 Years
Thinness: 5-9 Years

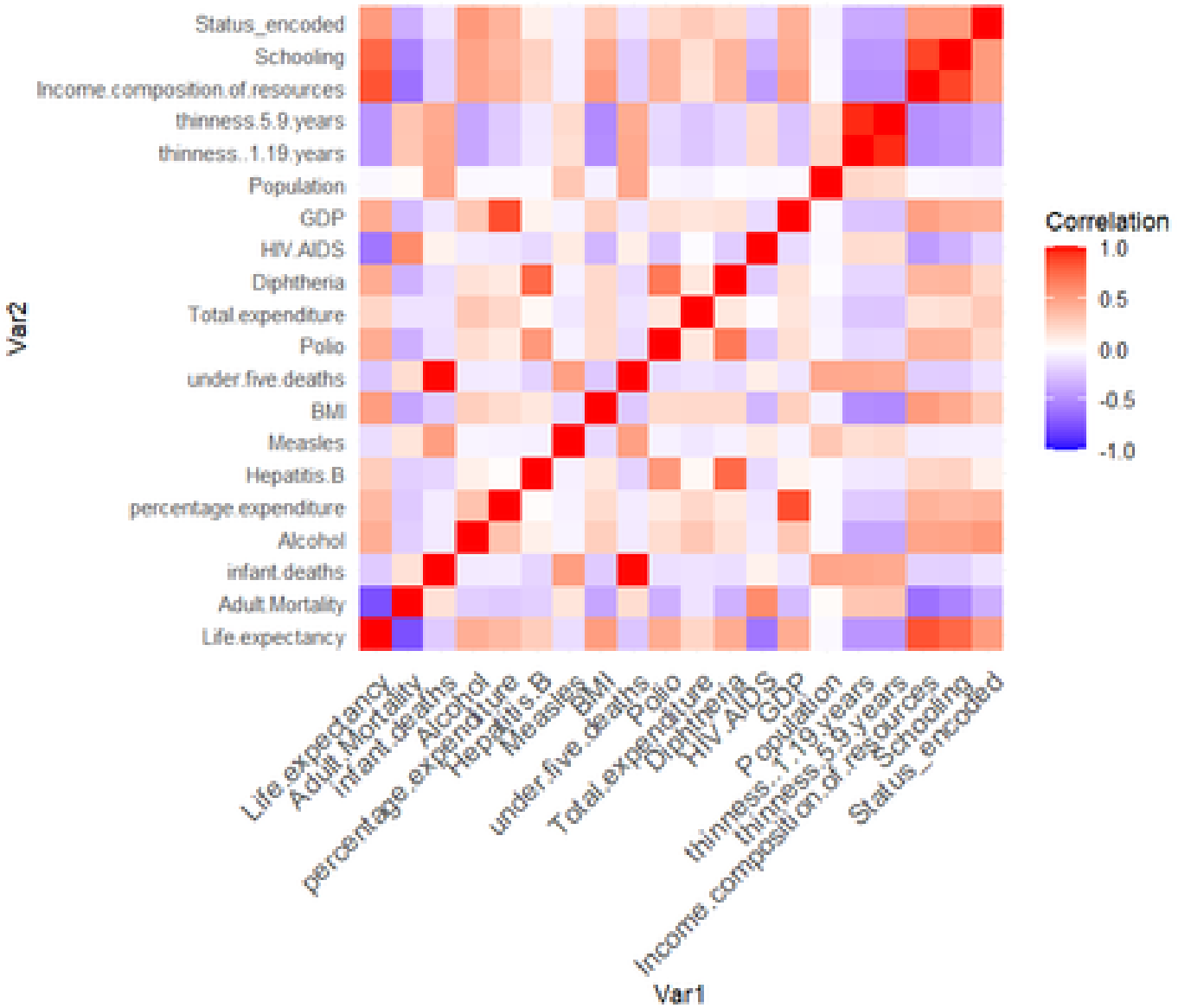
Exploratory Data Analysis

Initial Cleanup

- Imputing missing values: Using Mean
- Selecting data between 2010-2015 to reduce the effect of changes over time
- Ordinal encoding of the binary feature ”Status”
  [Developing = 0, Developed = 1]
- Ignoring the ”Countries” variable as many indicators of socioeconomic status are already considered

Exploratory Data Analysis

Visualizing Relationships



Exploratory Data Analysis

Visualizing Relationships

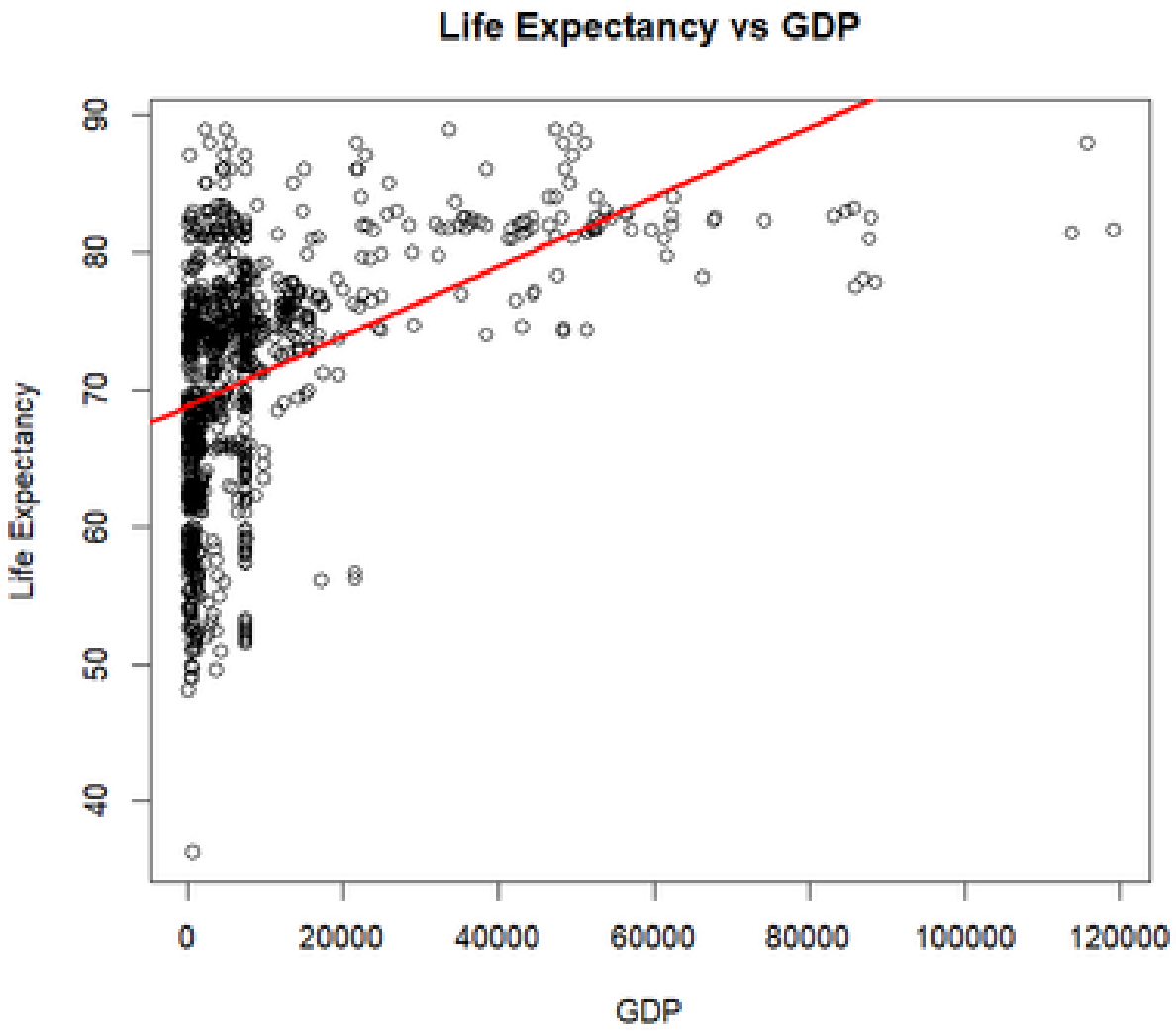


Histogram of the dependant variable

Simple Linear Regression

Simple Linear Regression

Life Expectancy vs GDP



Simple Linear Regression

Life Expectancy vs GDP

The linear regression equation obtained is:

Life Expectancy = 68.81 + 0.0002532 * GDP

i.e, for every 10,000 dollars increase in per capita GDP, the life expectancy increases by about 2.5 years

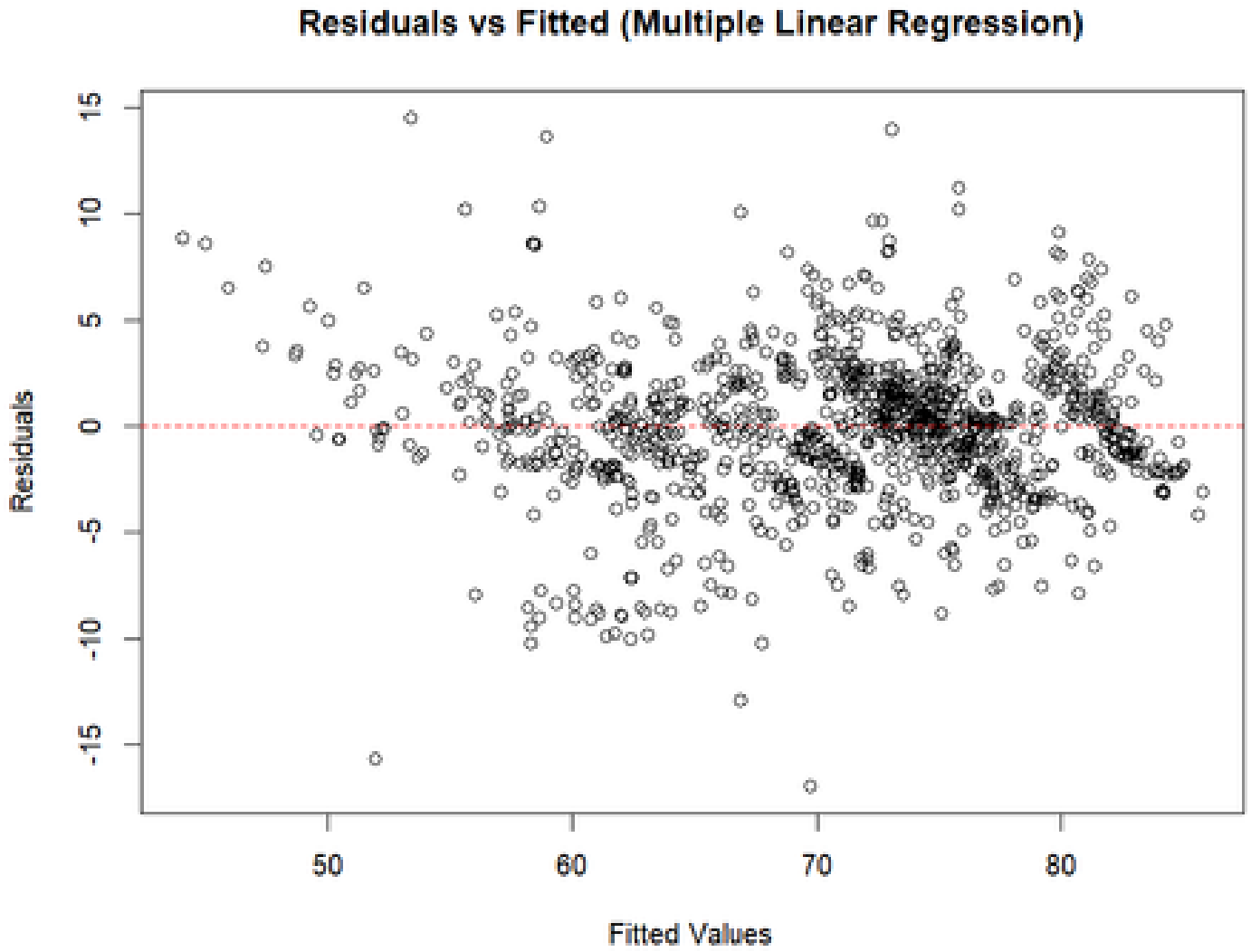
- The intercept is estimated to be 68.81 with a standard error of 0.2706, and the t-value is 254.31 with a p-value less than 2.2e-16, indicating that it is significantly different from zero.

- The coefficient of GDP is estimated to be 0.0002532 with a standard error of 1.589e-05, and the t-value is 15.94 with a p-value less than 2.2e-16, indicating that it is significantly different from zero.

The adjusted R-squared value is 0.1861, signifying that approximately 18.61% of the variance in life expectancy can be explained by GDP. The residual standard error is 7.773, and the F-statistic is 254.1, with a corresponding p-value of less than 2.2e-16. These results suggest that GDP is a significant predictor of life expectancy.

Multiple Linear Regression

Multiple Linear Regression



Multiple Linear Regression

We conducted a multiple linear regression analysis to predict life expectancy.

The model includes various factors such as Adult Mortality, Alcohol, Hepatitis B, BMI, Polio, Diphtheria, GDP, thinness (1-19 years), Income Composition of Resources, Infant Deaths, Percentage Expenditure, Measles, Under-five Deaths, Total Expenditure, HIV/AIDS, Population, thinness (5-9 years), Schooling, and Status.

The model resulted in an adjusted R-squared of 0.8364, indicating that approximately 83.64% of the variability in life expectancy can be explained by the independent variables.

Key predictors of life expectancy include **Adult Mortality** ($p < 2e\text{-}16$), **Income Composition of Resources** ($p < 2e\text{-}16$), **HIV/AIDS** ($p < 2e\text{-}16$), and **Status** ($p = 1.51e\text{-}05$).

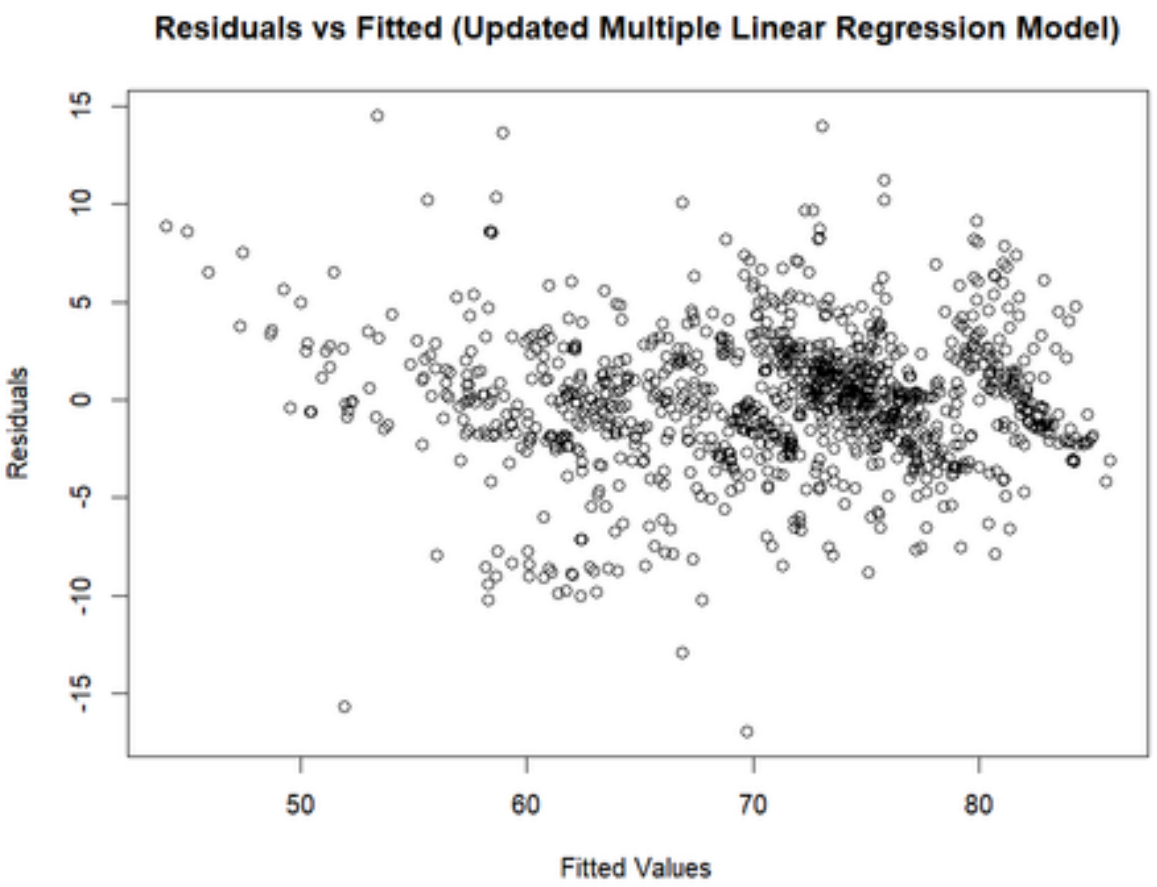
Further analysis is recommended to explore and address the non-significant predictors and the potential collinearity issue involving Hepatitis B, Polio, and thinness (1-19 years).

Stepwise Regression and AIC

Based on the Akaike Information Criterion (AIC) values and the stepwise regression, the variables were selected for the model as follows:

- At Step 1, thinness 5-9 years was removed, resulting in an AIC value of 2784.5.
- At Step 2, thinness 1-19 years was removed, yielding an AIC of 2784.8.
- After Step 3, percentage expenditure was removed, with an AIC of 2784.8.
- After Step 4, Population was removed, with an AIC of 2784.8.
- After Step 5, GDP was removed, with an AIC of 2785.0.
- The process was then stopped as removing further variables led to an increase in AIC value

Updated Model

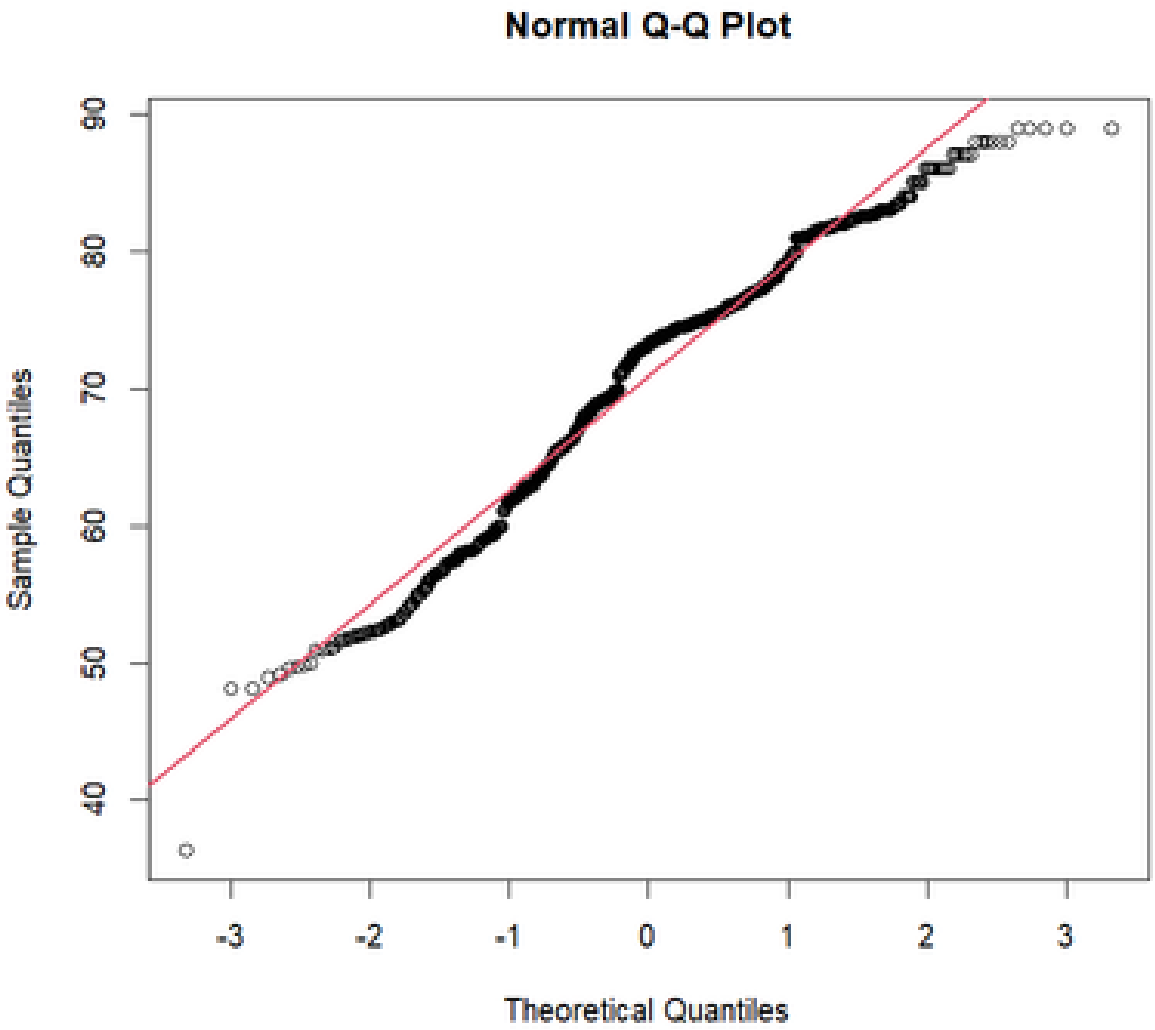


MSE = 12.12749

- The multiple R-squared value of 0.8388 suggests that 83.88% of the variance in life expectancy can be explained by the selected variables.
- The coefficients indicate that Adult Mortality, Income composition of resources, and HIV/AIDS have the most significant impact on life expectancy, followed by under-five deaths, Diphtheria, and Status.
- Measles and Hepatitis B did not show any statistical significance.
- The Residual standard error of 3.482 indicates the average difference between the observed and predicted values, suggesting that the model has a decent degree of predictive accuracy.

Regression Diagnostics

Checking normality



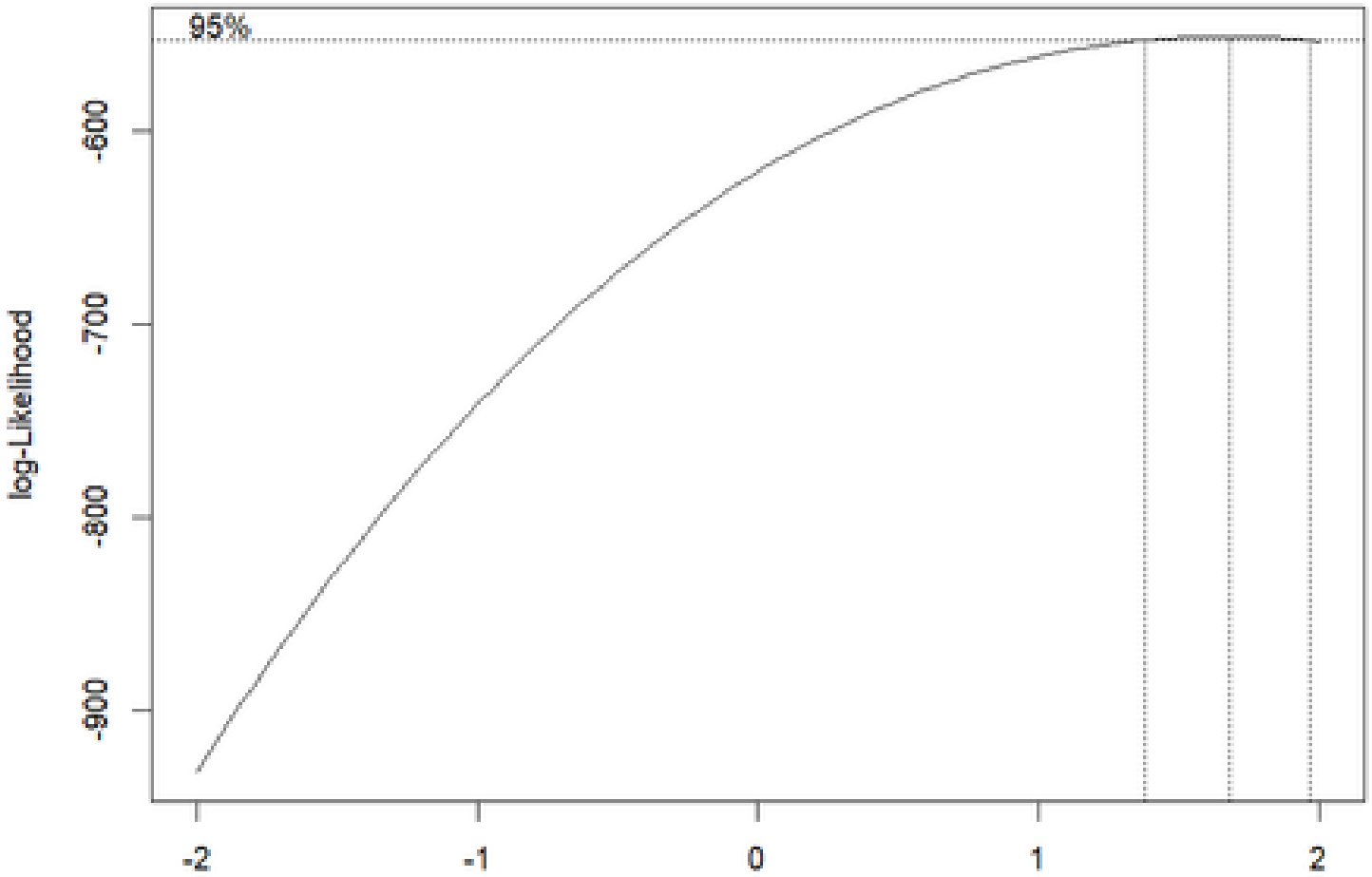
Checking Normality of the dependant variable

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        Shapiro-Wilk normality test

data:  df$Life.expectancy
W = 0.97031, p-value = 2.774e-14
```

Therefore, we have sufficient evidence to conclude that the life expectancy data is not normally distributed.

Transformations: Boxcox



Upon using the boxcox function, we found that the lambda value for the best transformation lies close to 2

Transformations: Square, Cube

- Upon doing square transformation, the p-value increased to 2.641e-10

However, the adjusted r-squared value for stepwise regression dropped to 0.8318

- Upon doing cube transformation, the p-value increased to 2.488e-09

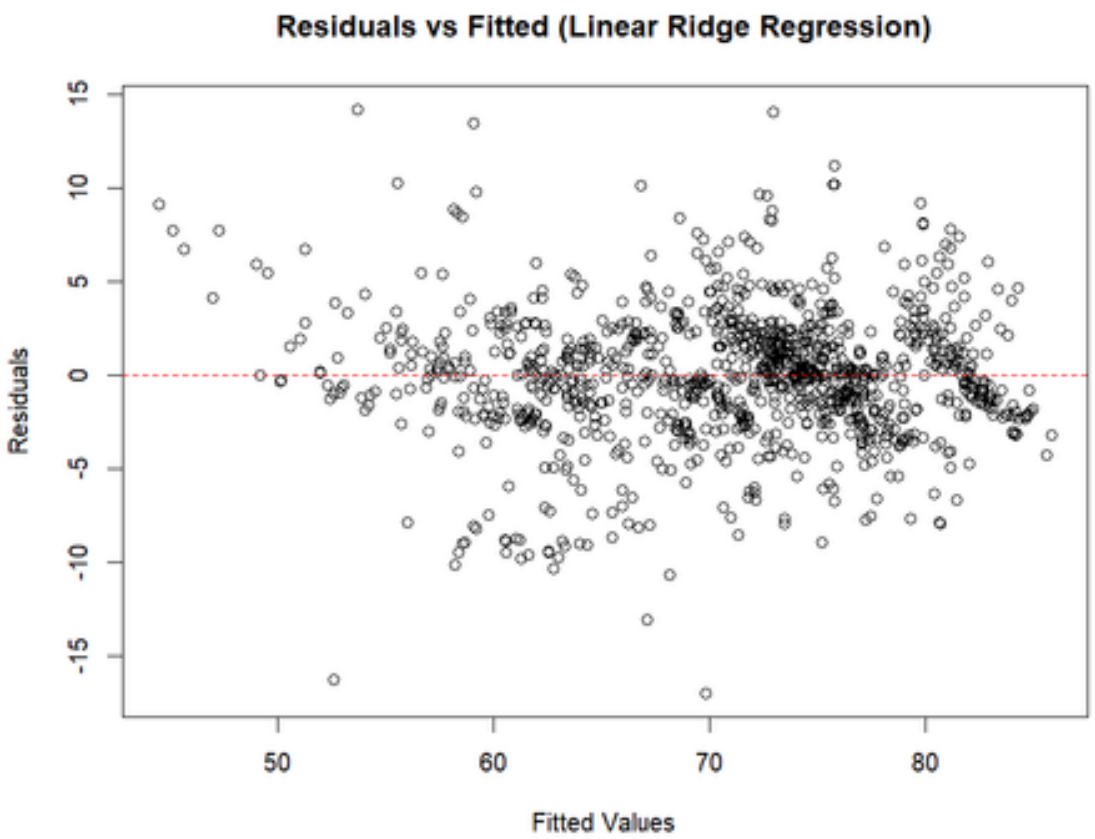
However, the adjusted r-squared value for stepwise regression dropped to 0.8366

Hence, we decided to stick to the original values

Diagnostics: Multicollinearity, Autocorrelation and Heteroscedasticity

- Variance Inflation Factor (VIF) analysis indicates significant multicollinearity between the variables infant.deaths (152.90) and under.five.deaths (148.89).
- The Durbin-Watson test revealed a significant positive autocorrelation (DW = 0.31, $p < 2.2e-16$), indicating the residuals are autocorrelated.
- The Goldfeld-Quandt test showed significant heteroscedasticity (GQ = 1.52, $p = 2.78e-12$), suggesting the variance changes across observations
- Further model improvement may be needed to address autocorrelation and heteroscedasticity issues.

Remedy: Linear Ridge Regression



MSE = 12.12749
R Square = 0.837551

- Ridge Parameter:

The ridge parameter comes out to 0.006807168.

- Degrees of freedom:
  - model 13.79
  - variance 13.58
  - residual 13.99

The model penalizes the coefficients only slightly.

Conclusion

The main factors affecting Life Expectancy for a country were found to be:

- Adult mortality,
- Income composition of resources,
- HIV/AIDS,
- Under five deaths,
- Diphtheria,
- Status

We can see that it is a mix of both Health as well as National Economic Indices that affect the life expectancy.